

## SIMATS ENGINEERING

### ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

#### COURSE OUTCOME COVERED

**CO2:** Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Scenario-Based Questions

#### Code of Conduct:

*I M.THANUJA(192425356)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.*

#### Signature of the student:

#### RUBRICS FOR CALCULATION:

| Criteria                     | Weightage | Level 4 (Exemplary)                                                       | Level 3 (Proficient)                               | Level 2 (Developing)                              | Level 1 (Beginning)                               | Marks (100) |
|------------------------------|-----------|---------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------|---------------------------------------------------|-------------|
| Understanding of Scenario    | 20%       | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps           | Partial understanding; misses key aspects         | Misinterprets or fails to understand the scenario |             |
| Identification of Key Issues | 20%       | Accurately identifies all relevant issues and factors involved            | Identifies most key issues with minor omissions    | Identifies limited issues; some irrelevant points | Unable to identify key issues                     |             |
| Application of Concepts      | 25%       | Applies appropriate concepts effectively to address the scenario          | Applies concepts with minor errors                 | Limited or partially correct application          | Unable to apply relevant concepts                 |             |
| Decision Making              | 20%       | Makes well-reasoned, logical decisions with strong rationale              | Makes reasonable decisions with some justification | Makes weak decisions with limited justification   | No clear decisions made or rationale provided     |             |
| Clarity of Response          | 15%       | Response is clear, structured, and logically presented                    | Generally clear with minor issues                  | Somewhat unclear or poorly structured             | Disorganized and difficult to understand          |             |

## Scenario-Based Questions

### **1. Low Contrast Image**

**A grayscale image appears dull with very little difference between object and background intensities.**

**(a) Clearly interpret the scenario and explain the cause of low contrast.**

A grayscale image has very little difference between the pixel intensities of the object and the background. As a result, the image appears dull, faded, and lacks visual clarity. Important structures are difficult to distinguish because the intensity values occupy only a narrow range of the available grayscale levels (0–255).

Low contrast usually occurs due to:

- Poor lighting conditions during image acquisition.
- Limited dynamic range of the camera sensor.
- Incorrect exposure settings.
- Atmospheric conditions such as fog or haze.
- Compression or transmission errors.

**(b) Identify all key issues affecting image visibility.**

The major issues include:

- Low separation between foreground and background.
- Loss of fine details and textures.
- Reduced edge visibility.
- Difficulty in detecting objects accurately.
- Poor segmentation performance.
- Reduced accuracy in feature extraction and image analysis.
- Human observers find it difficult to interpret the image.

**(c) Apply an appropriate enhancement technique to address the problem.**

Histogram Equalization

Histogram Equalization redistributes the intensity values over the complete grayscale range, thereby increasing the overall contrast.

Steps:

1. Calculate the histogram.
2. Compute cumulative distribution function (CDF).
3. Normalize the CDF.

4. Map old pixel values to new intensity values.
5. Generate the enhanced image.

**(d) Justify your choice with logical reasoning.**

Histogram Equalization is selected because:

- Automatically enhances overall contrast.
- Makes hidden details visible.
- Improves edge visibility.
- Enhances segmentation accuracy.
- Does not require manual parameter adjustment.
- Computationally efficient.
- Suitable for medical images, satellite images, and scanned documents.

Compared to simple brightness adjustment, Histogram Equalization improves the entire intensity distribution rather than shifting all pixels equally.

**(e) Present your explanation in a clear and structured manner.**

Problem: Narrow intensity range causing poor visibility.

Cause: Poor illumination, limited dynamic range, incorrect exposure.

Solution: Histogram Equalization.

Result:

- Better contrast.
- Enhanced object visibility.
- Improved edge detection.
- Better image interpretation.

## **2. Uneven Illumination**

**An image captured under non-uniform lighting shows bright and dark patches.**

**(a) Interpret the scenario and explain the impact of uneven illumination.**

The image contains bright and dark regions because illumination is not uniform across the scene. Some areas receive excessive light while others remain under exposed. This causes inconsistent brightness, making object detection difficult.

**(b) Identify key factors causing intensity variation.**

Factors Causing Intensity Variation

Major causes include:

- Uneven lighting sources.
- Shadows.
- Reflective surfaces.
- Camera sensor non-uniformity.
- Lens vignetting.
- Improper exposure.
- Environmental lighting conditions.

**(c)Apply a suitable enhancement method to normalize illumination.**

**Adaptive Histogram Equalization (CLAHE)**

Contrast Limited Adaptive Histogram Equalization (CLAHE) divides the image into small regions and enhances each region independently.

Procedure

1. Divide image into tiles.
2. Apply histogram equalization.
3. Limit excessive contrast amplification.
4. Interpolate neighboring regions.
5. Produce balanced illumination.

**(d)Justify your selected approach with reasoning.**

CLAHE is preferred because:

- Corrects local illumination.
- Prevents over-enhancement.
- Preserves details.
- Reduces noise amplification.
- Produces natural-looking images.
- Suitable for X-ray, MRI, fingerprint, and satellite images.

Unlike ordinary Histogram Equalization, CLAHE enhances each local region independently.

**(e)Provide a clear and well-structured explanation.**

Structured Explanation

Problem: Uneven lighting.

Cause: Shadows, non-uniform light, sensor limitations.

Technique: CLAHE.

Outcome:

- Balanced brightness.
- Improved local contrast.

- Better feature visibility.
- Enhanced segmentation.

### **3.Salt-and-Pepper Noise**

**A scanned document contains salt-and-pepper noise affecting readability.**

**(a)Explain the scenario and characteristics of this noise.**

Scenario Interpretation

Salt-and-Pepper Noise appears as randomly distributed white (255) and black (0) pixels.

It commonly occurs during:

- Image transmission
- Sensor malfunction
- Scanning errors
- Memory corruption

The document becomes difficult to read because random pixels replace original information.

**(b)Identify key issues impacting image quality.**

- Missing characters.
- Broken text.
- Reduced readability.
- False edges.
- Poor OCR accuracy.
- Incorrect segmentation.

**(c)Apply an appropriate filtering technique.**

Median Filter

The Median Filter replaces each pixel with the median intensity value of its neighboring pixels.

Procedure

1. Select a window (3×3 or 5×5).
2. Sort neighbouring pixels.
3. Find the median.
4. Replace centre pixel.
5. Repeat.

**(d)Justify why this method is more suitable than alternatives.**

Median Filter is preferred because:

- Removes impulse noise effectively.
- Preserves edges.
- Does not blur text.
- Simple implementation.
- Better than Mean Filter for impulse noise.

Mean filtering averages pixels, causing blurred characters, while Median filtering preserves sharp boundaries.

### **(e)Present your response clearly and logically.**

Problem: Impulse noise.

Cause: Scanner or transmission errors.

Technique: Median Filter.

Benefits:

- Removes black and white noise.
- Preserves text.
- Improves OCR accuracy.
- Maintains edge sharpness.

## **3. Blurred Edges**

**An image processed for noise removal appears overly smooth, causing edges to blur.**

### **(a)Interpret the scenario and explain why edges are blurred.**

After applying strong smoothing filters, the image becomes visually smooth but important edges disappear.

Edges contain high-frequency information, which smoothing filters often remove.

### **(b) Identify key issues related to smoothing and edge loss.**

- Loss of edge information.
- Reduced object boundaries.
- Poor segmentation.
- Difficulty in feature extraction.
- Lower recognition accuracy.

**(c)Apply an appropriate technique to restore edges.**

Unsharp Masking

Unsharp Masking sharpens edges by adding high-frequency information back into the image.

Steps

1. Blur original image.

2. Compute

$$\text{Mask} = \text{Original} - \text{Blurred}$$

mask:

3. Add

mask

back:

$$\text{Enhanced} = \text{Original} + k \times \text{Mask}$$

**(d)Justify your decision with clear reasoning.**

Advantages include:

- Restores edges.
- Enhances fine details.
- Improves object boundaries.
- Adjustable sharpening.
- Simple implementation.

Compared to Laplacian sharpening, Unsharp Masking introduces fewer artifacts and produces more natural results.

**(e)Structure your explanation clearly.**

Problem: Over-smoothing.

Cause: Excessive low-pass filtering.

Technique: Unsharp Masking.

Result

- Sharper edges.
- Improved details.
- Better feature extraction.
- Enhanced segmentation.

**4.High-Frequency Noise**

**A satellite image contains fine-grained noise affecting feature visibility.**

**(a)Interpret the scenario and describe high-frequency noise.**

Problem: Over-smoothing.

Cause: Excessive low-pass filtering.

Technique: Unsharp Masking.

Result

- Sharper edges.
- Improved details.
- Better feature extraction.
- Enhanced segmentation.

**(b)Identify the key issues affecting image clarity.**

- Poor feature visibility.
- Reduced image quality.
- Incorrect texture analysis.
- Difficult edge detection.
- Reduced classification accuracy.

**(c)Apply an appropriate frequency domain filtering method.**

Gaussian Low-Pass Filter (GLPF)

The image is transformed into the frequency domain using the Fourier Transform.

High-frequency noise is suppressed using a Gaussian Low-Pass Filter.

Finally, the inverse Fourier Transform reconstructs the enhanced image.

Procedure

1. Apply FFT.
2. Design Gaussian LPF.
3. Multiply filter with frequency spectrum.
4. Remove high-frequency components.
5. Apply Inverse FFT.

**(d)Justify the choice of filter with reasoning.**

GLPF is selected because:

- Smoothly removes noise.
- Minimizes ringing artifacts.
- Preserves important image structures.
- Suitable for satellite and remote sensing images.
- Better than Ideal Low-Pass Filter, which creates ringing due to abrupt frequency cut-off.

**(e)Present your answer in a clear and organized manner.**



Problem: High-frequency noise.

Cause: Sensor interference and atmospheric effects.

Technique: Gaussian Low-Pass Filter.

Benefits

- Noise reduction.
- Better image clarity.
- Improved feature visibility.
- Higher classification accuracy.